Tanner O’Rourke

Machine Learning Engineer

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5+ years of production software engineering experience and formal ML training through UT Austin's MS in AI. Bridges research and deployment, translating ML models into production-ready solutions with measurable business impact.

TECHNICAL SKILLS

ML/AI: PyTorch (+ TorchVision, GeoTorch), Scikit-learn, Hugging Face Transformers, XGBoost, NLP, Deep Learning, Neural Networks, Model Training/Evaluation, Feature Engineering
Languages: Python, SQL, JavaScript/TypeScript, Java
Data & Infrastructure: Pandas, NumPy, xarray, Git, CI/CD, AWS S3
Core Competencies: ML Pipeline Development, Model Evaluation, Data Analysis, Cross-functional Leadership

EXPERIENCE

Lead Engineer | DirecTV

October 2023 – August 2025 | *Promoted from Software Engineer in 3 months*

- Optimized /channel-lineup (2nd highest-traffic page) with SSR refactor: 49% faster LCP, 37% smaller bundle, 93% faster interactivity (INP), near-zero layout shift.
- Led site-wide WCAG 2.2 compliance initiative, remediating 200+ violations and mitigating $1M+ legal exposure.
- Modernized shared component library with CI/CD automation, enabling all .com teams to manage package versions independently.
- Built component testing GUI (self-initiated) for lower-environment previews without stage deploys.
- Awarded Q3 Delivery Excellence - top individual contributor across all .com teams.

Software Engineer | Warner Bros.

October 2020 – October 2023

- Sole owner of requirements, design, and development for 7 enterprise apps: casting payments, script management, production assets.
- Reduced data entry time by ~2× through unified interface design; validated via user testing.
- Directed offshore team on architecture, UI, and code quality—contributed directly to codebase to unblock progress.

EDUCATION

Master of Science: Artificial Intelligence | UT Austin | 2025-2026
Focus: Applied ML systems, generative modeling, human-centered AI

Bachelor of Science: Computer Science | CU Boulder | 2016-2020 | Chancellor’s Scholar
Minor: Psychology | Focus: Human-Computer Interaction

Relevant Coursework: Deep Learning, Natural Language Processing, Machine Learning, Reinforcement Learning, Deep Generative Models, AI in Healthcare

ML PROJECTS

FireFusion: ConvFormer-based spatiotemporal model for wildfire ignition prediction. Integrated 25+ features across meteorological, fuel, topographic, and human-risk modalities from 10 data sources. Implemented physics-aware masking and dual-head decoder for ignition probability and cause. Achieved 0.62 AUPRC vs. 0.08 baseline (7.75×) on WA test set. *Tech: PyTorch, xarray, MODIS, gridMET.* [Code] [Paper]

Mental Health Symptom Trajectory: End-to-end EHR feature pipeline: 30+ patient-day features from Synthea Coherent data including PHQ-9 trajectories, PDC-90 medication adherence, utilization patterns, and sparse bag-of-codes event streams. Trained multi-head XGBoost classifier for 30-day prediction of non-adherence, relapse, and deterioration. [Code] [Paper]

Lexical Negation Robustness: Analyzed SNLI for negation artifacts across 7 syntactic categories. Fine-tuned ELECTRA-small with targeted augmentation and slice-aware reweighting; improved accuracy on premise+hypothesis negation examples from 50% to 84.6%. [Code] [Paper]